

JEE MAIN REVISION MASTER: ISOLATION OF ELEMENTS OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 106 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Isolation Of Elements - Core Application	106 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Isolation Of Elements - Core Application (Very High Yield — 106 PYQs)

Core Principle: Key problem pattern in Isolation Of Elements.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2025 Evening Shift] Q#1 (MCQ): "The purification method based on the following physical transformation is : $\text{X(s)} \xrightarrow{\text{Heat}} \text{Y(g)}$ " [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : In the Ellingham diagram,..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Which of the following is used as a stabilizer during the concentration of sulphide ores?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 28th January Evening Shift (Q#1)

The purification method based on the following physical transformation is : $\text{Solid} \xrightarrow{\leq} \text{Vapour} \xrightarrow{\leq} \text{Solid}$

- (A) Extraction
- (B) Crystallization
- (C) Distillation
- (D) Sublimation

Correct Answer: None

Solution Analysis: The purification method described by the transformation: $\text{Solid} \xrightarrow{\leq} \text{Vapour} \xrightarrow{\leq} \text{Solid}$ is known as Sublimation . In sublimation, a solid substance directly transforms into its gaseous phase upon heating and reverts back to the solid phase upon cooling. This technique is commonly used to purify substances, particularly those that sublime more easily than others under the same conditions. Sublimation is effective for separating substances with diff